Lab 2: Linux Commands - II

Execute the following Linux commands on Linux terminal and verify the output.

NOTE- write command in lowercase only

1. Ifconfig - Display network interfaces and IP addresses.

```
harsh@Ubuntu:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::1fbc:b8c8:401:5df9  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:79:13:85  txqueuelen 1000  (Ethernet)
        RX packets 298253  bytes 377118018 (377.1 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 70966  bytes 14051767 (14.0 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 16267  bytes 1819152 (1.8 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 16267  bytes 1819152 (1.8 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

2. Kill - Kill active processes by process ID or name

```
harsh@Ubuntu:~$ kill -l
 1) SIGHUP       2) SIGINT       3) SIGQUIT      4) SIGILL       5) SIGTRAP
 6) SIGABRT      7) SIGBUS       8) SIGFPE       9) SIGKILL     10) SIGUSR1
11) SIGSEGV     12) SIGUSR2     13) SIGPIPE     14) SIGALRM     15) SIGTERM
16) SIGSTKFLT   17) SIGCHLD     18) SIGCONT     19) SIGSTOP     20) SIGTSTP
21) SIGTTIN     22) SIGTTOU     23) SIGURG      24) SIGXCPU     25) SIGXFSZ
26) SIGVTALRM   27) SIGPROF     28) SIGWINCH    29) SIGIO       30) SIGPWR
31) SIGSYS      34) SIGRTMIN    35) SIGRTMIN+1  36) SIGRTMIN+2  37) SIGRTMIN+3
38) SIGRTMIN+4  39) SIGRTMIN+5  40) SIGRTMIN+6  41) SIGRTMIN+7  42) SIGRTMIN+8
43) SIGRTMIN+9  44) SIGRTMIN+10 45) SIGRTMIN+11 46) SIGRTMIN+12 47) SIGRTMIN+13
48) SIGRTMIN+14 49) SIGRTMIN+15 50) SIGRTMAX-14 51) SIGRTMAX-13 52) SIGRTMAX-12
53) SIGRTMAX-11 54) SIGRTMAX-10 55) SIGRTMAX-9  56) SIGRTMAX-8  57) SIGRTMAX-7
58) SIGRTMAX-6  59) SIGRTMAX-5  60) SIGRTMAX-4  61) SIGRTMAX-3  62) SIGRTMAX-2
63) SIGRTMAX-1  64) SIGRTMAX
```

3. Mount - Mount file systems in Linux

```
harsh@Ubuntu:~$ mount
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,relatime)
udev on /dev type devtmpfs (rw,nosuid,relatime,size=1973768k,nr_inodes=493442,mo
de=755,inode64)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=620,ptmxmod
e=000)
tmpfs on /run type tmpfs (rw,nosuid,nodev,noexec,relatime,size=401824k,mode=755,
inode64)
/dev/sda3 on / type ext4 (rw,relatime,errors=remount-ro)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relat
ime)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,inode64)
tmpfs on /run/lock type tmpfs (rw,nosuid,nodev,noexec,relatime,size=5120k,inode6
4)
```

```
harsh@Ubuntu:~$ mount -V
mount from util-linux 2.37.2 (libmount 2.37.2: selinux, smack, btrfs, verity, na
mespaces, assert, debug)
```

4. Sort - Linux command to sort the content of a file while outputting

```
harsh@Ubuntu:~/Desktop/g11$ cat file1.txt
c cat
z zebra
l lion
d dog
b buffalo
p parrot
harsh@Ubuntu:~/Desktop/g11$ sort file1.txt
b buffalo
c cat
d dog
l lion
p parrot
z zebra
```

5. **Export -** Export environment variables in Linux

```
harsh@Ubuntu:~/Desktop/g11$ export -p
declare -x COLORTERM="truecolor"
declare -x DBUS_SESSION_BUS_ADDRESS="unix:path=/run/user/1000/bus"
declare -x DESKTOP_SESSION="ubuntu"
declare -x DISPLAY=":0"
declare -x GDMSESSION="ubuntu"
declare -x GNOME_DESKTOP_SESSION_ID="this-is-deprecated"
declare -x GNOME_SETUP_DISPLAY=":1"
declare -x GNOME_SHELL_SESSION_MODE="ubuntu"
declare -x GNOME_TERMINAL_SCREEN="/org/gnome/Terminal/screen/23110a04_1016_43aa_9943_4e
7c129a61fd"
declare -x GNOME_TERMINAL_SERVICE=":1.99"
declare -x GTK_MODULES="gail:atk-bridge"
declare -x HOME="/home/harsh"
declare -x IM_CONFIG_PHASE="1"
declare -x LANG="en_IN"
declare -x LANGUAGE="en_IN:en"
declare -x LESSCLOSE="/usr/bin/lesspipe %s %s"
declare -x LESSOPEN="| /usr/bin/lesspipe %s"
declare -x LOGNAME="harsh"
```

6. **Ssh -** Secure Shell command in Linux

```
harsh@Ubuntu:~/Desktop/g11$ ssh -l 192.168.1.1
usage: ssh [-46AaCfGgKkMNnqsTtVvXxYy] [-B bind_interface]
           [-b bind_address] [-c cipher_spec] [-D [bind_address:]port]
           [-E log_file] [-e escape_char] [-F configfile] [-I pkcs11]
           [-i identity_file] [-J [user@]host[:port]] [-L address]
           [-l login_name] [-m mac_spec] [-O ctl_cmd] [-o option] [-p port]
           [-Q query_option] [-R address] [-S ctl_path] [-W host:port]
           [-w local_tun[:remote_tun]] destination [command [argument ...]]
```

7. **Zip -** Zip files in Linux

```
harsh@Ubuntu:~/Desktop/g11$ ls
file1.txt  file2.txt
harsh@Ubuntu:~/Desktop/g11$ zip test.zip file1.txt file2.txt
  adding: file1.txt (stored 0%)
  adding: file2.txt (stored 0%)
harsh@Ubuntu:~/Desktop/g11$ ls
file1.txt  file2.txt  test.zip
```

8. **Unzip -** Unzip files in Linux

```
harsh@Ubuntu:~/Desktop/g11$ unzip test.zip
Archive:  test.zip
replace file1.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
 extracting: file1.txt
 extracting: file2.txt
harsh@Ubuntu:~/Desktop/g11$ ls
file1.txt  file2.txt  test.zip
```

9. **Ps-** Display active processes

```
harsh@Ubuntu:~/Desktop/g11$ ps
    PID TTY          TIME CMD
   3413 pts/0    00:00:00 bash
   3444 pts/0    00:00:00 ps
```

10. **Uname -** Linux command to get basic information about the OS

```
harsh@Ubuntu:~/Desktop/g11$ uname
Linux
```

11. **Chown -** Command for granting ownership of files or folders

```
harsh@Ubuntu:~/Desktop/g11$ chown harsh file1.txt
harsh@Ubuntu:~/Desktop/g11$ ls -l file1.txt
-rw-rw-r-- 1 harsh harsh 46 Jan 16 20:53 file1.txt
```

12. **Wget -** Direct download files from the internet

```
harsh@Ubuntu:~/Desktop/g11$ wget https://sot.pdpu.ac.in/downloads/UG_Curriculum_
CSED_2020-24.pdf
--2023-01-16 22:36:38--  https://sot.pdpu.ac.in/downloads/UG_Curriculum_CSED_202
0-24.pdf
Resolving sot.pdpu.ac.in (sot.pdpu.ac.in)... 203.77.200.33
Connecting to sot.pdpu.ac.in (sot.pdpu.ac.in)|203.77.200.33|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3799836 (3.6M) [application/pdf]
Saving to: 'UG_Curriculum_CSED_2020-24.pdf'

UG_Curriculum_CSED_20 100%[=======================>]   3.62M  1.19MB/s    in 3.
0s

2023-01-16 22:36:42 (1.19 MB/s) - 'UG_Curriculum_CSED_2020-24.pdf' saved [379983
6/3799836]
```

13. **Ufw -** Firewall command

```
root@Ubuntu:/home/harsh# sudo ufw status verbose
Status: inactive
root@Ubuntu:/home/harsh# sudo ufw enable
Firewall is active and enabled on system startup
root@Ubuntu:/home/harsh# sudo ufw disable
Firewall stopped and disabled on system startup
```

14. **Traceroute -** Trace all the network hops to reach the destination

```
root@Ubuntu:/home/harsh# traceroute www.pdpu.ac.in
traceroute to www.pdpu.ac.in (203.77.200.33), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  3.291 ms  3.222 ms  3.140 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  *^C
```

15. Service - Linux command to start and stop services

```
harsh@Ubuntu:~/Desktop$ service --status-all
 [ + ]  acpid
 [ - ]  alsa-utils
 [ - ]  anacron
 [ + ]  apparmor
 [ + ]  apport
 [ + ]  avahi-daemon
 [ - ]  bluetooth
 [ - ]  console-setup.sh
 [ + ]  cron
 [ + ]  cups
 [ + ]  cups-browsed
 [ + ]  dbus
 [ + ]  gdm3
 [ - ]  grub-common
 [ - ]  hwclock.sh
 [ + ]  irqbalance
 [ + ]  kerneloops
 [ - ]  keyboard-setup.sh
 [ + ]  kmod
 [ - ]  open-vm-tools
 [ + ]  openvpn
 [ - ]  plymouth
 [ + ]  plymouth-log
 [ + ]  procps
 [ - ]  pulseaudio-enable-autospawn
 [ - ]  rsync
 [ - ]  saned
 [ - ]  speech-dispatcher
 [ - ]  spice-vdagent
 [ + ]  udev
 [ + ]  ufw
 [ + ]  unattended-upgrades
 [ - ]  uuidd
 [ - ]  whoopsie
 [ - ]  x11-common
```

16. Alias - Create custom shortcuts for your regularly used commands

```
harsh@Ubuntu:~/Desktop$ alias c="clear"
harsh@Ubuntu:~/Desktop$ alias
alias alert='notify-send --urgency=low -i "$([ $? = 0 ] && echo terminal || echo
 error)" "$(history|tail -n1|sed -e '\''s/^\s*[0-9]\+\s*//;s/[;&|]\s*alert$//'\'
')"'
alias c='clear'
alias egrep='egrep --color=auto'
alias fgrep='fgrep --color=auto'
alias grep='grep --color=auto'
alias l='ls -CF'
alias la='ls -A'
alias ll='ls -alF'
alias ls='ls --color=auto'
```

17. Dd - Majorly used for creating bootable USB sticks

```
harsh@Ubuntu:~/Desktop/g11$ touch abc.txt xyz.txt
harsh@Ubuntu:~/Desktop/g11$ ls
abc.txt  file2.txt  test.zip  UG_Curriculum_CSED_2020-24.pdf  xyz.txt
harsh@Ubuntu:~/Desktop/g11$ echo "This is abc.txt" >> abc.txt
harsh@Ubuntu:~/Desktop/g11$ echo "This is xyz.txt" >> xyz.txt
harsh@Ubuntu:~/Desktop/g11$ cat abc.txt xyz.txt
This is abc.txt
This is xyz.txt
harsh@Ubuntu:~/Desktop/g11$ dd if=abc.txt of=xyz.txt
0+1 records in
0+1 records out
16 bytes copied, 0.000778132 s, 20.6 kB/s
harsh@Ubuntu:~/Desktop/g11$ cat xyz.txt
This is abc.txt
```

18. Whereis - Locate the binary, source, and manual pages for a command

```
harsh@Ubuntu:~/Desktop$ whereis bash
bash: /usr/bin/bash /usr/share/man/man1/bash.1.gz
```

19. Whatis - Find what a command is used for

```
harsh@Ubuntu:~/Desktop$ whatis cp rm touch
cp (1)               - copy files and directories
rm (1)               - remove files or directories
touch (1)            - change file timestamps
```

20. Diff - Find the difference between two files

```
harsh@Ubuntu:~/Desktop/g11$ diff file1.txt file2.txt
1,6d0
< c cat
< z zebra
< l lion
< d dog
< b buffalo
< p parrot
```

21. Ln - Create symbolic links (shortcuts) to other files

```
harsh@Ubuntu:~/Desktop/g11$ ln -s file1.txt link.txt
harsh@Ubuntu:~/Desktop/g11$ ls -l link.txt
lrwxrwxrwx 1 harsh harsh 9 Jan 17 19:26 link.txt -> file1.txt
```

22. Top - View active processes live with their system usage

```
harsh@Ubuntu:~/Desktop/g11$ top

top - 19:27:39 up 24 min,  1 user,  load average: 0.00, 0.00, 0.03
Tasks: 185 total,   1 running, 184 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.7 us,  0.7 sy,  0.0 ni, 98.2 id,  0.2 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   3924.0 total,   2477.1 free,    671.5 used,    775.5 buff/cache
MiB Swap:   2680.0 total,   2680.0 free,      0.0 used.   3007.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1446 harsh     20   0 3954140 309284 125224 S   3.0   7.7   0:55.39 gnome-s+
    280 root     -51   0       0      0      0 S   0.3   0.0   0:01.09 irq/18-+
   1377 harsh     39  19  642080  28164  18776 S   0.3   0.7   0:01.26 tracker+
   1957 harsh     20   0  581280  61368  41844 S   0.3   1.5   0:11.63 gnome-t+
      1 root      20   0  166712  11860   8284 S   0.0   0.3   0:01.93 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.02 kthreadd
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_par+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_fl+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker+
      9 root      20   0       0      0      0 I   0.0   0.0   0:03.15 kworker+
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_perc+
```

23. Useradd - Add new user or change existing users data

```
root@Ubuntu:/home/harsh/Desktop/g11# useradd harsh2
Password:
```

24. Man - Access manual pages for all Linux commands

```
root@Ubuntu:/home/harsh/Desktop/g11# man -f sleep
sleep (1)            - delay for a specified amount of time
sleep (3)            - sleep for a specified number of seconds
```

25. Cp - Similar usage as mv but for copying files in Linux

```
harsh@Ubuntu:~/Desktop/g11$ cat file2.txt
This is file2.txt
harsh@Ubuntu:~/Desktop/g11$ cp file1.txt file2.txt
harsh@Ubuntu:~/Desktop/g11$ cat file2.txt
c cat
z zebra
l lion
d dog
b buffalo
p parrot
```

26. Ln - Create symbolic links (shortcuts) to other files

```
harsh@Ubuntu:~/Desktop/g11$ ln -s file1.txt link.txt
harsh@Ubuntu:~/Desktop/g11$ ls -l link.txt
lrwxrwxrwx 1 harsh harsh 9 Jan 17 19:26 link.txt -> file1.txt
```

27. Netstat - netstat command displays various network related information such as network connections, routing tables, interface statistics, masquerade connections, multicast memberships etc.

```
harsh@Ubuntu:~/Desktop/g11$ netstat -nr
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
0.0.0.0         10.0.2.2        0.0.0.0         UG        0 0          0 enp0s3
10.0.2.0        0.0.0.0         255.255.255.0   U         0 0          0 enp0s3
169.254.0.0     0.0.0.0         255.255.0.0     U         0 0          0 enp0s3
```

28. **Nslookup -** A network utility program used to obtain information about Internet servers. As its name suggests, the utility finds name server information for domains by querying DNS.

```
harsh@Ubuntu:~/Desktop/g11$ nslookup yahoo.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   yahoo.com
Address: 98.137.11.163
Name:   yahoo.com
Address: 74.6.143.26
Name:   yahoo.com
Address: 74.6.143.25
Name:   yahoo.com
Address: 74.6.231.20
Name:   yahoo.com
Address: 74.6.231.21
Name:   yahoo.com
Address: 98.137.11.164
Name:   yahoo.com
Address: 2001:4998:44:3507::8001
Name:   yahoo.com
Address: 2001:4998:124:1507::f001
Name:   yahoo.com
Address: 2001:4998:24:120d::1:0
Name:   yahoo.com
Address: 2001:4998:44:3507::8000
Name:   yahoo.com
Address: 2001:4998:24:120d::1:1
Name:   yahoo.com
```

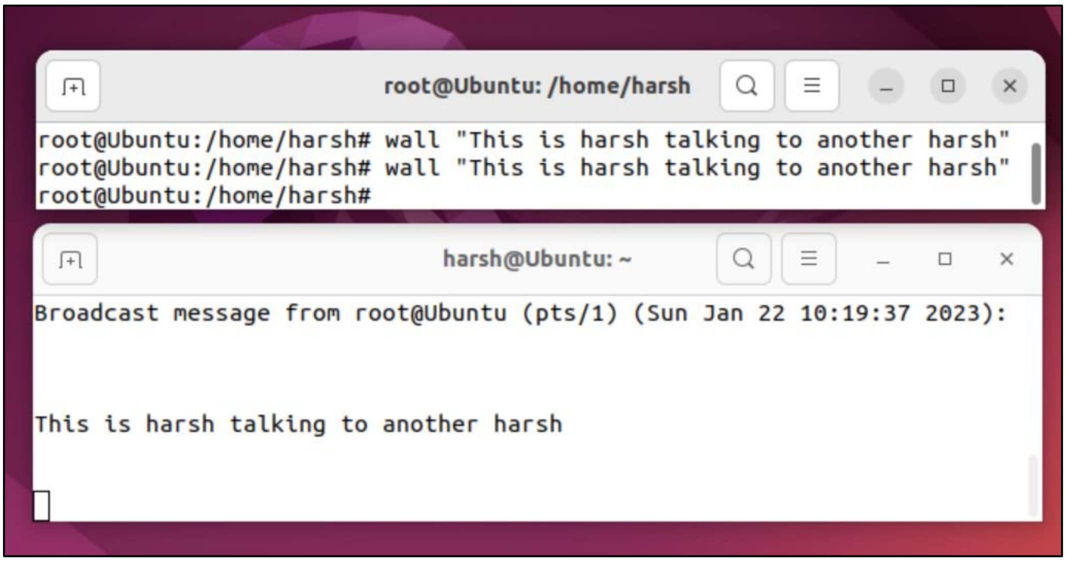
29. **Dig -** dig is a tool for querying DNS nameservers for information about host addresses, mail exchanges, nameservers, and related information. This tool can be used from any Linux (Unix) or Macintosh OS X operating system. The most typical use of dig is to simply query a single host.

```
harsh@Ubuntu:~$ dig google.com

; <<>> DiG 9.18.1-1ubuntu1.1-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 33320
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                    IN      A

;; ANSWER SECTION:
google.com.             0       IN      A       142.251.42.14

;; Query time: 12 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jan 19 10:52:59 IST 2023
;; MSG SIZE  rcvd: 55
```

30. Uptime - You have just connected to your Linux Server Machine and founds Something unusual or malicious, what you will do? Guessing…. NO, definitely not you could run uptime to verify what happened actually when the server was unattended.

```
harsh@Ubuntu:~$ uptime
 10:54:49 up 12 min,  1 user,  load average: 2.10, 1.38, 0.80
```

31. Wall - one of the most important command for administrator, wall sends a message to everybody logged in with their mesg permission set to "yes". The message can be given as an argument to wall, or it can be sent to wall's standard input.



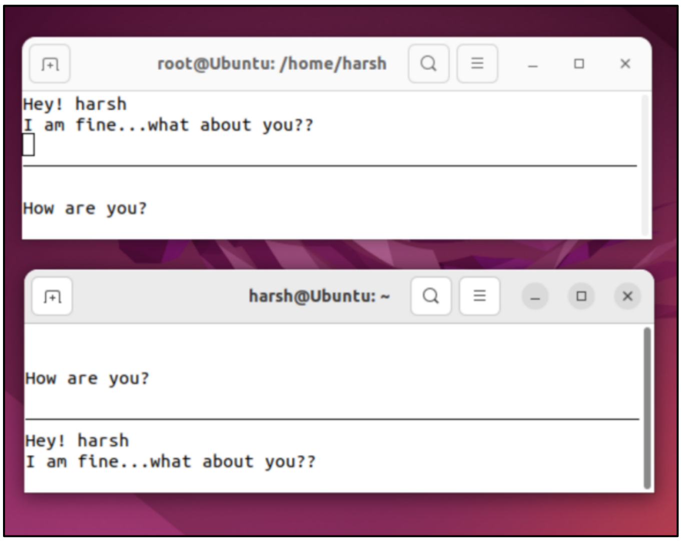
32. Mesg - Lets you control if people can use the "write" command, to send text to you over the screen.

```
harsh@Ubuntu:~/Desktop/g11$ mesg
is y
harsh@Ubuntu:~/Desktop/g11$ mesg n
harsh@Ubuntu:~/Desktop/g11$ mesg
is n
```

33. Write - Let you send text directly to the screen of another Linux machine if 'mesg' is 'y'.

```
root@Ubuntu:/home/harsh# write harsh
Hey Harsh! Break in 30 mins
Regards, HSroot@Ubuntu:/home/harsh# write harsh
Hey Harsh! Break in 30 mins
Regards, HS
root@Ubuntu:/home/harsh# w
 11:26:10 up 44 min,  3 users,  load average: 0.59, 0.63, 0.83
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU WHAT
harsh    tty2     tty2             10:43   44:36   0.07s  0.07s /usr/libexec/gn
harsh    pts/1    -                11:18    1.00s  0.18s  0.07s sudo login hars
harsh    pts/1    -                11:19    1.00s  0.18s  0.06s -bash
```

34. Talk - An enhancement to write command, talk command lets you talk to the logged in users.



35. **W -** what command 'w' seems you funny? But actually it is not. t's a command, even if it's just one letter long! The command "w" is a combination of uptime and who commands given one immediately after the other, in that order.

```
root@Ubuntu:/home/harsh# w
 11:33:07 up 51 min,  3 users,  load average: 3.95, 1.91, 1.25
USER     TTY      FROM             LOGIN@   IDLE   JCPU   PCPU WHAT
harsh    tty2     tty2             10:43   51:33  0.07s  0.07s /usr/libexec/gn
harsh    pts/1    -                11:18    1.00s  0.41s  0.11s sudo login hars
harsh    pts/1    -                11:19    1.00s  0.41s  0.06s -bash
```

36. **Rename -** As the name suggests, this command rename files. rename will rename the specified files by replacing the first occurrence from the file name.

```
harsh@Ubuntu:~/Desktop/g11$ cat file1.txt
c cat
z zebra
l lion
d dog
b buffalo
p parrot
harsh@Ubuntu:~/Desktop/g11$ cat file2.txt
This is file2.txt

harsh@Ubuntu:~/Desktop/g11$ mv file1.txt file2.txt
harsh@Ubuntu:~/Desktop/g11$ cat file1.txt
cat: file1.txt: No such file or directory
harsh@Ubuntu:~/Desktop/g11$ cat file2.txt
c cat
z zebra
l lion
d dog
b buffalo
p parrot
```

37. **Free -** Keeping track of memory and resources is as much important, as any other task performed by an administrator, and 'free' command comes to rescue here.

```
harsh@Ubuntu:~/Desktop/g11$ free
               total        used        free      shared  buff/cache   available
Mem:         4018208     1413360      683316       55136     1921532     2318144
Swap:        2744316           0     2744316
```